Intro to Deep Q-Learning

By Luca Ianniello

Summary of Reinforcement Learning

- **What is Reinforcement Learning?**
  - An area of machine learning where an agent learns to make decisions by interacting with an environment to maximize a reward.
- **Key Components**:
  - Agent
  - Environment
  - State
  - Action
  - Reward
  - Policy

Core Idea of RL: Learn a policy that maps states to optimal actions to maximize cumulative reward.

Summary of Reinforcement Learning

- **Value Function**:
  - Measures the expected cumulative reward starting from state and following a particular policy.

$$v_\pi(s) = \mathbb{E}_\pi[G_t | S_t = s] = \mathbb{E}_\pi\left[\sum_{k=0}^{T-t+1} \gamma^k R_{t+k+1} \middle| S_t = s\right]$$

- **Action-Value Function**:
  - Measures the expected cumulative reward of taking action in state and following a particular policy thereafter.

$$q_\pi(s, a) = \mathbb{E}_\pi[G_t | S_t = s, A = a] = \mathbb{E}_\pi\left[\sum_{k=0}^{T-t+1} \gamma^k R_{t+k+1} \middle| S_t = s, A_t = a\right]$$

$$v_*(s) = \max_a q_*(s, a)$$

What is Q-Learning?

- Q-Learning is a value-based Reinforcement Learning (RL) algorithm that seeks to learn the optimal action-selection policy for an agent interacting with an environment.
- It is an **Off-Policy Algorithm**:
  - It learns the optimal policy independently of the agent’s current behavior policy.
- It is a **Model-Free Algorithm**:
  - It does not require knowledge of the environment’s dynamics

$$Q(S, A) \leftarrow Q(S, A) + \alpha \left(R + \gamma \max_{a'} Q(S', a') - Q(S, A) \right)$$

What is Deep Q-Learning?

- Why Deep Q-Learning?
  - Q-Learning works well only in small, discrete state-action spaces.
  - In traditional Q-Learning, the Q-values are stored in a table or matrix.
- In Deep Q-Learning, a neural network takes the current state as input and outputs predicted Q-values for all possible actions. The weights of the network (denoted as) are updated during training to minimize the error in Q-value predictions.
- Key Innovation:
  - **Experience Replay**
  - **Target Network**
  - **Loss function**

$$\mathcal{L}_i(w_i) = \mathbb{E}_{s,a,r,s' \sim \mathcal{D}_i}\left[\left(r + \gamma \max_{a'} Q(s',a';w_i^-) - Q(s,a;w_i)\right)^2\right]$$

Applications and Advantages of DQN

- Applications:
  - Video Games:
    - Deep Q-Learning has been famously used to achieve superhuman performance in Atari games by learning directly from raw pixel inputs.
  - Robotics:
    - Used for navigation, control, and decision-making tasks in robotic systems.
  - Finance:
    - Applied in portfolio optimization and trading strategies.
- Advantages:
  - Handles high-dimensional, continuous state spaces.
  - Learns directly from raw sensory inputs like images or sensor data.
  - Scales to complex, real-world problems.

Recap and Key messages

- Classical Q-Learning is a foundational RL algorithm but is limited by its inability to scale to high-dimensional problems.
- Deep Q-Learning overcomes these challenges by leveraging neural networks to approximate the Q-function.
- Innovations like experience replay and target networks are critical for stable training and efficient learning.
- Deep Q-Learning has opened the door to solving complex, real-world tasks in gaming, robotics, and beyond.

Thanks for the attention!